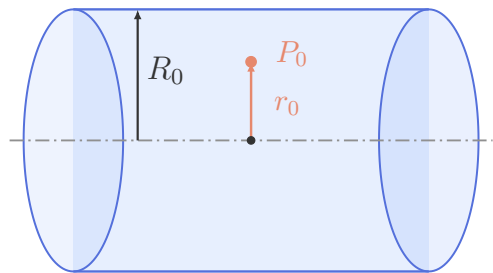


初始状态 $t = 0$

物理坐标 $0 \leq r \leq R_0$



$L = 0.25 \text{ m}$ (不变)

比例径向收缩

$$v(r, t) = \frac{R'(t)}{R(t)} r$$

$$r(t) = \frac{R(t)}{R_0} r_0$$

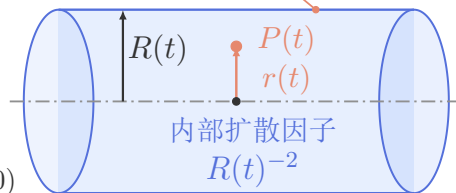
$$R'(t) < 0, \quad v(r, t) < 0 \quad (r > 0)$$



收缩后 $t > 0$

材料坐标 $\xi = r/R(t) \in [0, 1]$

表面梯度因子 $R(t)^{-1}$



$L = 0.25 \text{ m}$ (不变)

同一材料点的相对位置不变

$$\xi = \frac{r_0}{R_0} = \frac{r(t)}{R(t)} = \text{const}$$

固定计算区间 $0 \leq \xi \leq 1$

$$\partial_r = \frac{1}{R(t)} \partial_\xi$$

$$\frac{1}{r} \partial_r (r D \partial_r) = \frac{1}{R^2(t) \xi} \partial_\xi (\xi D \partial_\xi)$$